

BRAINDOUGH: A Local-First Neuroencoding Harness with a BOLD5000 Metadata-Control Benchmark

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Abstract

Large public visual fMRI datasets make it practical to test neuroencoding workflows against measured brain data before investing in heavyweight model features or bespoke stimulus pipelines. This preprint describes a local-first extension of BRAINDOUGH for BOLD5000 Release 1.0 and reports the most informative current result: a small source-family metadata signal that recurs across subjects, ROIs, timing runs, and split variants. The benchmark uses ridge regression on coarse stimulus provenance features, not image pixels or deep neural features, and is therefore intended as a metadata-control, provenance-content baseline, and harness-validation task rather than as a competitive neural encoding model. Across four BOLD5000 participants and six visual ROIs, the primary source-family rows produced mean Pearson $r = 0.0748$ and mean $R^2 = 0.00244$, with the strongest effects in PPA and LOC rather than early visual cortex. In the full TR34 run, the best source-family row was CSI2 LHPPA ($r = 0.154$, $R^2 = 0.0161$), with a run-level max-statistic permutation $p = 0.000999$ over the searched subject, ROI, and model grid. The canonical primary summary uses TR34 for CSI1–CSI3 and TR3 for CSI4. A grouped-filename sensitivity analysis eliminated exact repeated-filename train/validation crossings and retained a positive best source-family result (CSI2 LHPPA, $r = 0.175$, $R^2 = 0.0317$, max-statistic $p = 0.00990$ with 100 permutations). These results support BRAINDOUGH as a reproducible measured-data evaluation harness and establish a concrete metadata baseline that future image-feature encoders can be evaluated against.

1 Introduction

The BOLD5000 dataset is a slow event-related fMRI study designed around large-scale natural image stimuli drawn from standard computer-vision image sources including SUN, COCO, and ImageNet [1]. This makes it useful not only for modeling visual cortex, but also for testing whether an analysis system can ingest real measured neuroimaging data, preserve dataset provenance, and report statistical checks before scaling to more expressive models.

BRAINDOUGH is a local-first experimental harness for brain-modeling experiments. Its design goal is practical reproducibility on a workstation: the repository contains code, tests, configs, and lightweight reports, while large datasets, model weights, caches, and run outputs live outside the checkout under a user-configured `BRAINDOUGH_HOME`. The current implementation includes fake backends for CI and measured-data adapters for BOLD5000 Release 1.0 ROI data.

The most interesting current result is intentionally modest. Coarse source-family metadata yields small positive validation correlations and low, sometimes near-zero, validation R^2 values, especially in LOC and PPA. This should not be read as evidence that visual cortex “encodes filenames” or that source-family metadata is a good encoding model. Instead, it is a useful baseline:

if a future image-feature model cannot beat this source/provenance-content control under the same split, it is not yet demonstrating much beyond broad image-source structure.

2 Dataset and Task

The benchmark uses BOLD5000 Release 1.0 processed ROI vectors and trial metadata. Release 1.0 is useful for a light, local benchmark because the processed ROI matrices are straightforward to run without reproducing the full fMRI preprocessing pipeline. The adapter records the release label, provenance, dataset terms, and a source caveat in each run manifest. The manuscript does not vendor raw BOLD5000 data or stimulus images.

Each run selects 1024 trials per subject using a fixed random seed (20260509). TR34 denotes the Release 1.0 combined TR3/TR4 ROI timing file. The primary timing is TR34 for CSI1–CSI3. CSI4 is reported from TR3 in the canonical primary summary because the BOLD5000 guidance recommends TR3 for CSI4 [3]; TR34 for CSI4 is retained only as timing sensitivity. All subjects use the same six ROI labels: left and right early visual cortex (LHEarlyVis, RHEarlyVis), LOC (LHLOC, RHLOC), and PPA (LHPPA, RHPPA).

Because BOLD5000 Release 2.0 provides updated functional estimates, including optimized GLM outputs, this Release 1.0 ROI benchmark should be interpreted as a local harness-validation and metadata-baseline analysis rather than as the definitive BOLD5000 encoding benchmark [3].

Three metadata models are evaluated per subject and ROI:

- **source_family**: one-hot stimulus source-family indicators plus a repeated-stimulus flag (four features in the Release 1.0 adapter);
- **token_hash**: hashed filename-token features used only as a leakage-stress diagnostic;
- **source_plus_token_hash**: concatenation of the two feature blocks, also diagnostic.

The primary scientific interpretation is restricted to **source_family**. Filename-token rows are reported by the harness so that repeated-stimulus and identifier effects are visible during development, but they are not treated as scientific encoding claims.

3 Methods

For each subject, ROI, and model family, the backend fits ridge regression [4] on a 75/25 split. Ridge alpha is selected from $\{0.1, 1, 10, 100\}$ by validation Pearson correlation. The same alpha-selection rule is replayed inside the row-wise and max-statistic permutation procedures. Because alpha is selected on the validation split, all scores are described as validation benchmark scores rather than as independent test estimates.

The default primary split stratifies by source family. A separate grouped-filename split stratifies by source while assigning exact repeated filenames to only one side of the train/validation boundary. This grouped split is used as a leakage-control sensitivity, not as a replacement for a full image-identity generalization study.

Model quality is reported as validation Pearson correlation and validation R^2 . Pearson r and R^2 are interpreted separately: small positive correlations can be detectable while R^2 remains near zero or negative. R^2 is therefore treated as the practical predictive-performance metric and Pearson r as a directional association metric. Bootstrap intervals are computed by resampling validation trials. Within-row permutation values are uncorrected diagnostics. Max-statistic values

are reported separately at the run level. Both use the largest positive Pearson statistic and are therefore one-sided diagnostics for positive predictive association. Permutation p -values use the finite-sample $(b + 1)/(n + 1)$ convention so that random-permutation p -values are never zero [5].

The code changes required for this benchmark are small and auditable: seeded random trial selection, subject/ROI-aware Release 1.0 indexing, split-balance diagnostics, repeated-filename leakage diagnostics, max-statistic permutations, and a repo-local Tectonic build target for this manuscript.

4 Results

4.1 Primary source-family benchmark

Table 1 summarizes the canonical primary result: CSI1–CSI3 use TR34, and CSI4 uses TR3. Across the 24 source-family rows, mean Pearson r is 0.0748, mean R^2 is 0.00244, and 14 of 24 rows have validation $R^2 > 0$. The effect is spatially structured: PPA has the largest average source-family score, followed by LOC, while early visual ROIs remain near zero by R^2 .

Table 1: Primary source-family results by ROI family. CSI1–CSI3 use TR34 and CSI4 uses TR3. Full row-level values are committed in `tables/primary_source_family_results.csv`.

Summary group	n	Mean r	Mean R^2	Rows with $R^2 > 0$
All primary rows	24	0.0748	0.00244	14
Early visual ROIs	8	0.0281	-0.00566	0
LOC ROIs	8	0.0760	0.00294	7
PPA ROIs	8	0.1204	0.01005	7

The best result in the primary TR34 all-subject run is CSI2 LHPPA under the source-family model ($r = 0.154$, $R^2 = 0.0161$, bootstrap interval $[0.111, 0.199]$). The max-statistic permutation value for the best observed TR34 grid result is $p = 0.000999$ with 1000 permutations. For CSI4 at TR3, the best source-family row is LHPPA ($r = 0.0820$, $R^2 = 0.00210$), and the full TR3 run also has a run-level max-statistic $p = 0.000999$. For both 1000-permutation runs, $p = 0.000999$ is the minimum finite value under the $(b + 1)/(n + 1)$ estimator. These values are detectable, but the practical effect sizes are small.

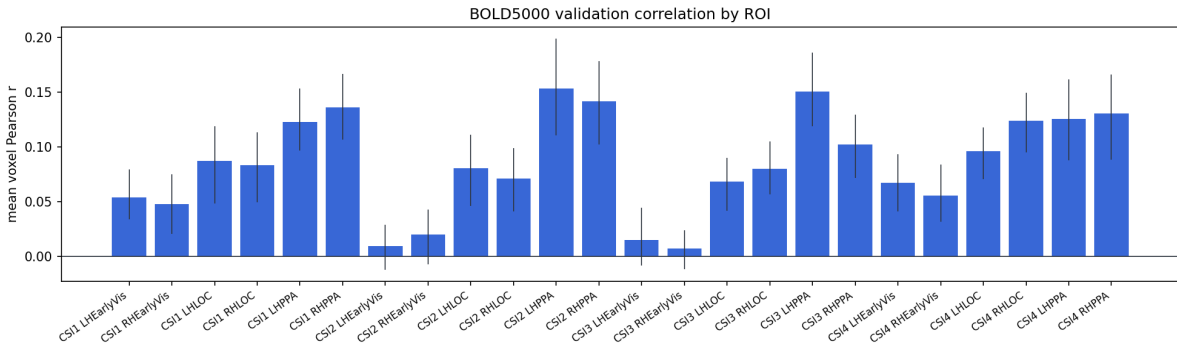


Figure 1: TR34 source-family and diagnostic model scores across subjects and ROIs. TR34 is the primary timing for CSI1–CSI3.

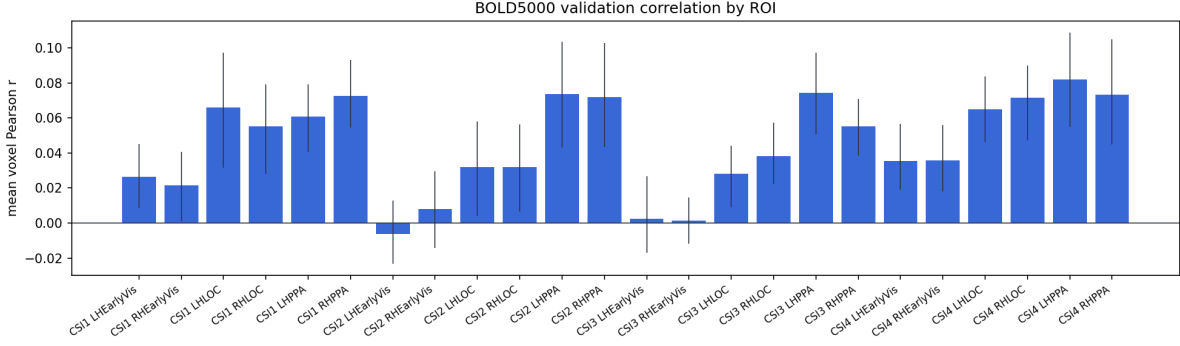


Figure 2: TR3 timing sensitivity. CSI4 TR3 rows are used in the canonical primary summary.

4.2 Repeated-filename sensitivity

The primary source-stratified split can allow exact filenames that occur more than once in the selected trials to appear on both sides of the train/validation boundary. This is not, by itself, evidence that the source-family model is leaking image identity: source family is a coarse provenance feature and does not include filename identity. It is nevertheless a useful stress point for a benchmark harness.

The grouped-filename sensitivity sets exact repeated-filename crossings to zero for every subject. The strongest source-family row remains CSI2 LHPPA ($r = 0.175$, $R^2 = 0.0317$, bootstrap interval $[0.123, 0.216]$), with a max-statistic value of $p = 0.00990$. Because this sensitivity used only 100 permutations, $p = 0.00990$ is the minimum finite value possible under the reported estimator and should be interpreted as zero exceedances among 100 shuffled max statistics rather than as a precisely estimated p -value. Across all 24 source-family rows in the grouped sensitivity, mean Pearson r is 0.0809 and mean R^2 is 0.00559.

The grouped split controls exact repeated-filename train/validation crossings; it does not eliminate broader source-, category-, session-, or image-distribution differences. Train/validation source-family counts and repeated-filename crossing counts are committed under `tables/` in the TR34 split-balance, TR34 leakage-diagnostic, grouped-TR34 split-balance, and grouped-TR34 leakage-diagnostic CSV files.

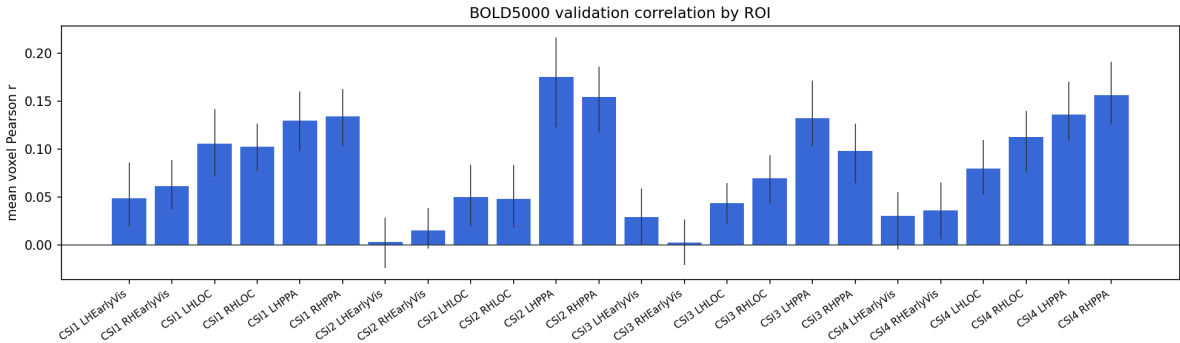


Figure 3: TR34 grouped-filename sensitivity. Exact repeated-filename train/validation crossings are eliminated while preserving the small source-family signal.

5 Discussion

The result is best interpreted as a measured-data harness check with scientific guardrails. The source-family signal recurs across subjects, ROIs, timing runs, and split variants, is strongest in high-level visual ROIs, and remains positive after a grouped-filename split that removes exact repeated-filename train/validation crossings. The same evidence also constrains the claim: explained variance is tiny, source-family labels are coarse, and the current adapter uses Release 1.0 ROI vectors rather than pixel-level image features, captions, model embeddings, voxelwise encoding, or Release 2.0 GLM outputs.

This makes the result useful precisely because it is limited. It establishes a low but measurable benchmark bar for future BRAINDOUGH encoders. A model using CLIP features, caption features, low-level image statistics, or learned visual embeddings should be expected to exceed this meta-data baseline if it captures image-level information beyond source-family structure under the same subject, ROI, trial, and split diagnostics. If it does not, the benchmark can catch that failure early.

6 Limitations

Several limitations are deliberate. First, this is not a state-of-the-art BOLD5000 analysis and does not estimate noise ceilings. Because BOLD5000 Release 2.0 provides updated functional estimates, including optimized GLM outputs, the Release 1.0 ROI benchmark here is a local harness-validation and metadata-baseline analysis rather than the definitive BOLD5000 encoding benchmark [3]. Second, the benchmark uses 1024 selected trials per subject rather than the full release. Third, the grouped-filename sensitivity controls exact repeated filenames but does not replace stronger image-identity, session, or category-generalization splits. Fourth, source-family labels may proxy broad visual and semantic differences among SUN, COCO, ImageNet, and other source groups; this is why the result is described as a provenance-content baseline rather than a mechanistic neural code. Fifth, the grouped sensitivity used 100 permutations for local runtime reasons, so its max-statistic value has coarse finite resolution. Sixth, permutation and bootstrap summaries are conditional on the selected BOLD5000 Release 1.0 trials and subjects. They are not population-level inference over human participants. Bootstrap intervals are validation-trial resampling intervals, not subject-level confidence intervals and not selection-adjusted intervals.

7 Reproducibility

The committed configs live in `experiments/local/` and are named for the TR34, TR3, and grouped-TR34 sensitivity runs.

The corresponding root commands are the three `run-bold5000-preprint-*` targets in the root `justfile`. Routine repository validation runs through `just ci`. This PDF is built with the `preprint-build` target against this timestamped directory.

8 Acknowledgments and AI Assistance

BOLD5000 was produced and released by Chang and colleagues [1]. ChatGPT 5.5 Extended Pro was used for iterative scientific critique, claim scoping, and drafting assistance. All numerical results reported here come from committed BRAINDOUGH configs and run artifacts, not from model-generated estimates.

A Run Summary

Table 2: Run-level summary. Full run IDs and notes are committed in `tables/run_summary.csv`.

Run	Split	Perms	Best ROI	Best r	Max-stat p
TR34 primary	stratified source	1000	CSI2 LHPPA	0.154	0.000999
TR3 timing	stratified source	1000	CSI4 LHPPA	0.082	0.000999
TR34 grouped sensitivity	grouped filename	100	CSI2 LHPPA	0.175	0.00990

B Canonical Primary Source-Family Rows

Table 3: Canonical primary source-family rows. CSI1–CSI3 use TR34 and CSI4 uses TR3. Within-row p values are uncorrected diagnostics.

Subject	TR	ROI	r	R^2	Within-row p	Bootstrap interval
CSI1	TR34	LHEarlyVis	0.054	-0.0006	0.000999	[0.034, 0.079]
CSI1	TR34	RHEarlyVis	0.048	-0.0028	0.000999	[0.020, 0.075]
CSI1	TR34	LHLOC	0.087	0.0041	0.000999	[0.048, 0.119]
CSI1	TR34	RHLOC	0.084	0.0057	0.000999	[0.050, 0.114]
CSI1	TR34	LHPPA	0.123	0.0109	0.000999	[0.097, 0.153]
CSI1	TR34	RHPPA	0.136	0.0149	0.000999	[0.107, 0.167]
CSI2	TR34	LHEarlyVis	0.010	-0.0074	0.168831	[-0.012, 0.029]
CSI2	TR34	RHEarlyVis	0.020	-0.0058	0.001998	[-0.007, 0.043]
CSI2	TR34	LHLOC	0.081	0.0027	0.000999	[0.046, 0.111]
CSI2	TR34	RHLOC	0.071	0.0025	0.000999	[0.041, 0.099]
CSI2	TR34	LHPPA	0.154	0.0161	0.000999	[0.111, 0.199]
CSI2	TR34	RHPPA	0.142	0.0148	0.000999	[0.103, 0.179]
CSI3	TR34	LHEarlyVis	0.015	-0.0096	0.041958	[-0.008, 0.044]
CSI3	TR34	RHEarlyVis	0.006	-0.0079	0.255744	[-0.013, 0.024]
CSI3	TR34	LHLOC	0.069	0.0037	0.000999	[0.042, 0.090]
CSI3	TR34	RHLOC	0.080	0.0048	0.000999	[0.057, 0.105]
CSI3	TR34	LHPPA	0.151	0.0161	0.000999	[0.119, 0.187]
CSI3	TR34	RHPPA	0.102	0.0067	0.000999	[0.072, 0.130]
CSI4	TR3	LHEarlyVis	0.036	-0.0046	0.024975	[0.019, 0.056]
CSI4	TR3	RHEarlyVis	0.036	-0.0067	0.003996	[0.018, 0.056]
CSI4	TR3	LHLOC	0.065	-0.0003	0.000999	[0.046, 0.084]
CSI4	TR3	RHLOC	0.071	0.0002	0.000999	[0.047, 0.090]
CSI4	TR3	LHPPA	0.082	0.0021	0.000999	[0.055, 0.109]
CSI4	TR3	RHPPA	0.073	-0.0014	0.000999	[0.045, 0.105]

References

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